

Michael (Spok) Szpakowicz

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Experience

Software Engineer

Ruffolo Nuclear Consulting: 2025 – 2026

- Created script for converting thousands of nuclear requirements into a custom reviewal pipeline.
- Set up a LLM development and deployment server for generating intent vectors for requirements.
- Wrote script for parsing data between SQL databases and various Microsoft Office files.

Lead Robotics R&D Developer

Admetal Robotics: 2023 – 2025

Guided and worked with a team of three on the following:

- Developing a player-following golf trolley with gesture control, ball tracking, and obstacle avoidance features using YOLOv11, a stereo camera, UWB, and an RK3588 chip.
- Fine-tuned and tested RDT-1B and OpenVLA on pick-place, sorting, and fine manipulation tasks for a 6-DoF arm and 2-finger gripper in order to deploy and train service robots in the real world.
- Built and tested UMI, DexCap and LeapHand systems for data collection and teleoperation.
- Built and programmed an autonomous mobile base with ROS2, a LiDAR, and a depth camera.
- Created an embodied autonomous assistant framework for one-armed service robots using ChatGPT, YOLOv8, and voice networks interfacing with a custom API and running on a Jetson AGX Xavier.

Robotics Software Engineer

Admetal Robotics: 2021 – 2023

- Developed a custom real-time human tracking solution on a Jetson Nano with YOLOv5, enabling a motorized golf caddy to follow a player while using minimal system resources.
- Created a camera-based part-locating program for robotic pick-and-place systems used by 3 of our clients.
- Implemented SLAM to allow a restaurant delivery robot to navigate indoor environments autonomously.

Self Driving Cars Specialization Instructor

Coursera: 2022 – 2023

- Verified and corrected course materials to enhance learning outcomes in *Introduction to Self Driving Cars*, a specialization taken by over 250,000 students since its release in 2019.

Teaching Assistant

University of Toronto: 2017 – 2020

- Built and configured 5 autonomous driving platforms for student use for *Intro to Mobile Robotics*.
- Led weekly tutorials for up to 40 students, marked tests and assignments within a week of their completion, and provided extended office hours on *Software Design* and *Computer Organization* courses.
- Answered student questions during *Intro to Programming* lectures to clarify any confusion in course content.

Set Animation Engineer

Little Canada: 2017 – 2018

- Implemented a programmable waterfall simulation using 5 LED strips and a microcontroller.
- Developed an Arduino library for complex PWM signal sequencing, resulting in 60% less programming.

Technical Skills

Robotics Tools: ROS(1&2), Gazebo, IsaacSim, PyBullet, CARLA, Python, C++, OpenCV, Linux, Jetson

ML Frameworks: PyTorch, Lightning, TensorFlow, Keras, TensorRT, RKNN, stable-baselines3

ML Networks: RDT, OpenVLA, Aloha, DexCap, UMI, DiT, PPO, HER, VAE, GPT, YOLO, CNN, RL

Education

M.Eng. Robotics Engineering

University of Toronto - 2023

H.B.Sc. Computer Science Specialist

University of Toronto - 2020

Projects and Hobbies

Graduate Research

Throwing Robot: 2023 - 2024

- Developed a PPO-based reinforcement learning framework for teaching throwing behaviour quickly and safely both in PyBullet, and in the real world on a mobile platform with a 6 DoF arm.

Undergraduate Research

Autonomous Car Projects: 2019 & 2020

- Worked with a team of 6 at the RVL Lab on 5 autonomous mobile platforms to study the influence of digital billboards on vision-based self driving cars through adversarial noise. Our convolutional neural networks showed promising results after significant neural network optimization.
- Collaborated with 2 other students to convert 3 toy cars into smartphone-based, indoor self-driving platforms. Multiple approaches were attempted but imitation learning proved best, and the cars were able to safely complete a loop through campus hallways.

Club Founder, Lead, and Mentor

UTM Robotics Club: 2019 – 2021

- Led weekly meetings on design, electronics, control systems, autonomous vehicles, and drones.
- Conducted workshops on Arduinos, 3D Printing, PID Loops, and Deep Learning.
- Created a self-balancing pendulum and implemented SLAM on MIT Racecars with ROS1.